No. 8 20.11.2024

$\boxed{1}$

$$y' = \frac{1}{2h}(y_{i+1} - y_{i-1}) = (d_1 y)_i \qquad y = y_i$$

$$y'' = \frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1}) = (D_{11} y)_i \quad (*) \qquad d_1 d_1 = D_{11} \ ?$$

$$(d_1 d_1 y)_i = d_1\left[\frac{1}{2h}(y_{i+1} - y_{i-1})\right]$$

$$= \frac{1}{2h}(d_1 y_{i+1} - d_1 y_{i-1})$$

$$= \frac{1}{2h}\left(\frac{1}{2h}(y_{i+2} - y_i) - \frac{1}{2h}(y_i - y_{i-2})\right)$$

$$= \frac{1}{4h^2}(y_{i+2} - 2y_i + y_{i-2})$$

$$d_1 d_1 y = \frac{1}{4h^2}(y_{i+2} - 2y_i + y_{i-2}) \qquad (**)$$

$$(d_0^+ y)_i = \frac{1}{2}(y_i + y_{i+1}), \quad (d_0^- y)_i = \frac{1}{2}(y_i + y_{i-1})$$

$$(d_1^+ y)_i = \frac{1}{h}(y_{i+1} - y_i), \quad (d_1^- y)_i = \frac{1}{h}(y_i - y_{i-1})$$

$$(D_{10} y)_i = (d_0^- d_1^+ y)_i = d_0^-\left(\frac{1}{h}(y_{i+1} - y_i)\right)$$

$$= \frac{1}{h}(d_0^- y_{i+1} - d_0^- y_i)$$

$$= \frac{1}{h}\left(\frac{1}{2}(y_{i+1} + y_i) - \frac{1}{2}(y_i + y_{i-1})\right)$$

$$= \frac{1}{2h}(y_{i+1} - y_{i-1})$$

$$(D_{01} y)_i = (d_1^- d_0^+ y)_i = \ldots\ldots = \frac{1}{2h}(y_{i+1} - y_{i-1})$$

$$(D_{11} y)_i = (d_1^+ d_1^- y)_i = d_1^+\left(\frac{1}{h}(y_i - y_{i-1})\right)$$

$$= \frac{1}{h}(d_1^+ y_i - d_1^+ y_{i-1})$$

$$= \frac{1}{h}\left[\frac{1}{h}(y_{i+1} - y_i) - \frac{1}{h}(y_i - y_{i-1})\right]$$

$$= \frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1})$$

$y' \approx \begin{cases} D_{10} \\ D_{01} \end{cases}$ $\qquad y'' \approx D_{11}$ $\qquad y \to y_i$

$y \to D_{00} = d_0^+ d_0^-$

$D_{00} D_{11} = D_{01} D_{10}$ (?)

$$D_{00} = d_0^+ d_0^- y_i = \frac{1}{4}(y_{i-1} + 2y_i + y_{i+1})$$

$$D_{01} D_{10} = D_{01} D_{01} = \frac{1}{4h^2}(y_{i+2} - 2y_i + y_{i-2})$$

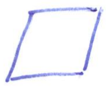
$$(D_{00} D_{11} y)_i = D_{00}\left(\frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1})\right)$$

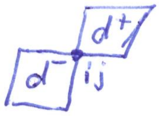
~~сох~~ суперпозиции двух первых производных

$$= \frac{1}{h^2}\left[\cancel{D_{00}}\,\frac{1}{4}(y_i + 2y_{i+1} + y_{i+2}) - \frac{2}{4}(y_{i-1} + 2y_i + y_{i+1}) + \frac{1}{4}(y_{i-2} + 2y_{i-1} + y_i)\right]$$

$$= \frac{1}{4h^2}\left[y_{i+2} - 2y_i + y_{i-2}\right]$$

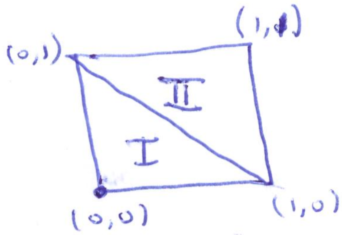
$$D_{00} D_{11} = d_0^+ d_0^- d_1^+ d_1^- = \underbrace{d_0^+ d_1^-}\ \underbrace{d_1^+ d_0^-} = D_{01} D_{10} \qquad \boxed{\frac{\partial^2}{\partial x^2} = \frac{d}{dx}\frac{d}{dx}}$$

 $\to D_{ij} = D_{ji}$ $\qquad D_{00} D_{11} = D_{01} D_{10}$



$d_m^+ = T_{11} d_m^-$

$$D_{ij} = d_i^+ d_j^- = T_{11} d_i^- d_j^- = T_{11} d_j^- d_i^- = d_j^+ d_i^- = D_{ji} \Rightarrow \boxed{D_{ij} = D_{ji}}$$

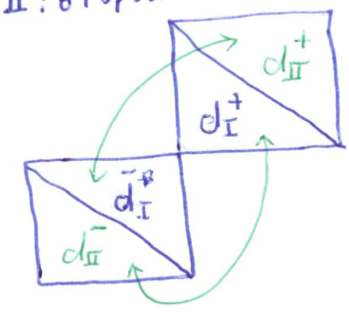


$\omega_1 = \{(0,0), (1,0), (0,1)\}$

$\omega_2 = \{(1,0), (1,1), (0,1)\}$

$\boxed{3}$ 1989

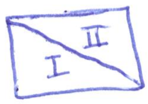
I: первый шаблон
II: второй шаблон



$$d^+_{m\,I} = T_{11}\, d^-_{m\,II} \quad , \quad d^+_{m\,II} = T_{11}\, d^-_{m\,I}$$

$$d^-_{m\,II} = T_{-1-1}\, d^+_{m\,I}$$

номер шаблона

~~Dij~~ 

$$D_{ij} \to \frac{1}{2}\left(D^{I}_{ij} + D^{II}_{ij}\right)$$

$$= \frac{1}{2}\left(d^+_{iI} d^-_{jI} + d^+_{iII} d^-_{jII}\right)$$

$$= \frac{1}{2}\left(d^+_{iI} d^-_{jI} + \cancel{T_{11}}\, d^-_{iI}\, \cancel{T_{-1-1}}\, d^+_{jI}\right)$$

$$= \frac{1}{2}\left(d^+_{iI} d^-_{jI} + d^+_{jI} d^-_{iI}\right)$$

$$= \frac{1}{2}\left(D^{I}_{ij} + D^{I}_{ji}\right)$$

$$\triangle \to D_{ij} \sim D^*_{ij} = \frac{1}{2}\left(D_{ij} + D_{ji}\right)$$

$$\left.\begin{matrix} \square \\ \boxed{\diagdown} \end{matrix}\right\} D_{ij} D_{kl} = D_{kj} D_{il} = D_{il} D_{kj}$$

$$\triangledown \ (?) \quad D^*_{ij} D^*_{kl} = D^*_{kj} D^*_{il} = D^*_{il} D^*_{kj} \qquad (D_{ij} \neq D_{ji})$$

~~$\frac{1}{2}\left(d^+_i d^-_j + d^+_j d^-_i\right) + \frac{1}{2}(T$~~

$$\frac{1}{2}\left(d^+_i d^-_j + d^+_j d^-_i\right)\frac{1}{2}\left(d^+_k d^-_i + d^+_i d^-_k\right) - \frac{1}{2}\left(d^+_k d^-_j + d^+_j d^-_k\right)\frac{1}{2}\left(d^+_i d^-_i + d^+_i d^-_i\right) =$$

$$= \frac{1}{4}\Big[d^+_i \cancel{d^-_j d^+_k} d^-_i + \underline{d^+_i d^-_j d^+_i d^-_k} + \underline{d^+_j d^-_i d^+_k d^-_i} + d^+_j \cancel{d^-_i d^+_i} d^-_k$$

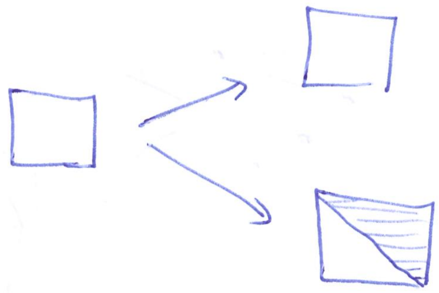
$$- d^+_k \cancel{d^-_j d^+_i} d^-_i - \underline{d^+_k d^-_j d^+_i d^-_i} - \underline{d^+_j d^-_k d^+_i d^-_i} - d^+_j \cancel{d^-_k d^+_i} d^-_i\Big]$$

$$= \frac{1}{4}\left[d^+_i d^-_k \left(d^+_i d^-_j - d^+_j d^-_i\right) + d^+_k d^-_i \left(d^+_j d^-_i - d^+_i d^-_j\right)\right]$$

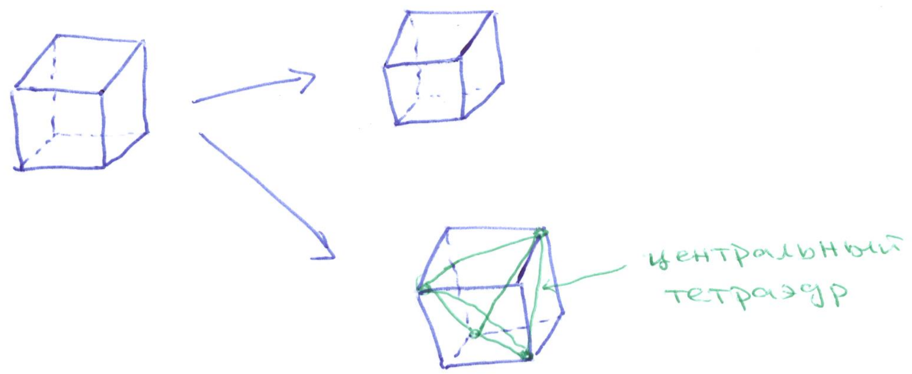
$$= \frac{1}{4}\left[\left(d^+_i d^-_j - d^+_j d^-_i\right)\left(d^+_i d^-_k - d^+_k d^-_i\right)\right] = \frac{1}{4}\left(D_{ij} - D_{ji}\right)\left(D_{ik} - D_{ki}\right)$$

$$\neq 0$$

2 D



3 D



$$\int_V \sigma_{ij}\,\delta\varepsilon_{ij}\,dV = \int_V \rho\,(F_i - \ddot{u}_i)\,\delta u_i\,dV + \int_{S_p} P_i\,\delta u_i\,dS$$

$P = \operatorname{div}\vec{u}$

div $\rightarrow$ $$(\lambda+\mu)\operatorname{grad}\operatorname{div}\bar{u} + \mu\,\Delta\bar{u} + \rho F = \rho\frac{\partial^2 u}{\partial t^2}$$

$\rightarrow (\lambda+2\mu)\,\Delta P + \rho\operatorname{div}F = \rho\dfrac{\partial^2 P}{\partial t^2}$

$\mu\,\Delta\varphi_i + \rho(\operatorname{rot}F)_i = \rho\dfrac{\partial^2\varphi_i}{\partial t^2}$

$$\begin{cases}(\lambda+\mu)\left(\dfrac{\partial^2 u_1}{\partial(x^1)^2} + \cancel{\dfrac{\partial^2 u_2}{\partial(x^1)^2}}\dfrac{\partial^2 u_2}{\partial x^1\partial x^2}\right) + \mu\left(\dfrac{\partial^2 u_1}{\partial(x^1)^2} + \dfrac{\partial^2 u_1}{\partial(x^2)^2}\right) + \rho F_1 = \rho\dfrac{\partial^2 u_1}{\partial t^2}\\[2ex] (\lambda+\mu)\left(\dfrac{\partial^2 u_1}{\partial x^1\partial x^2} + \dfrac{\partial^2 u_2}{\partial(x^2)^2}\right) + \mu\left(\dfrac{\partial^2 u_2}{\partial(x^1)^2} + \dfrac{\partial^2 u_2}{\partial(x^2)^2}\right) + \rho F_2 = \rho\dfrac{\partial^2 u_2}{\partial t^2}\end{cases}$$

$$\begin{cases}(\lambda+\mu)\,(D_{11}u_1 + D_{12}u_2) + \mu\,D_\Delta u_1 + \rho F_1 = \rho D_{tt}u_1\\ (\lambda+\mu)\,(D_{21}u_1 + D_{22}u_2) + \mu\,D_\Delta u_2 + \rho F_2 = \rho D_{tt}u_2\end{cases}$$

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$$\begin{cases} D_{20} \to (\lambda+\mu)(D_{11}u_1 + D_{12}u_2) + \mu D_\Delta u_1 + \rho F_1 = \rho D_{tt} u_1 \leftarrow D_{01} \\ - \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad + \\ D_{10} \to (\lambda+\mu)(D_{21}u_1 + D_{22}u_2) + \mu D_\Delta u_2 + \rho F_2 = \rho D_{tt} u_2 \leftarrow D_{02} \end{cases}$$

$\operatorname{div} \bar{u} = D_{01}\bar{u}_1 + D_{02}u_2$

$$(\lambda+\mu)\left[D_{11}D_{01}u_1 + D_{11}D_{02}u_2 + D_{22}D_{01}u_1 + D_{22}D_{02}u_2\right] + \mu D_\Delta (D_{01}u_1 + D_{02}u_2) + \rho(D_{01}F_1 + D_{02}F_2) = \rho D_{tt}(D_{01}u_1 + D_{02}u_2)$$

$$[\cdots] = D_{11}(D_{01}u_1 + D_{02}u_2) + D_{22}(D_{01}u_1 + D_{02}u_2) = D_\Delta(D_{01}u_1 + D_{02}u_2)$$

$$(\lambda+2\mu) D_\Delta P + \rho(D_{01}F + D_{02}F) = \rho D_{tt}\Delta P$$

$$(\lambda+\mu)\left[D_{11}\cancel{D_{20}}u_1 + D_{12}\cancel{D_{20}}u_2 - D_{21}\cancel{D_{10}}u_1 - D_{22}\cancel{D_{10}}u_2\right]$$

$$\mu D_\Delta q + \rho(D_{20}F_1 - D_{10}F_2) = \rho D_{tt} q$$

$q = D_{20}u_1 - D_{10}u_2$